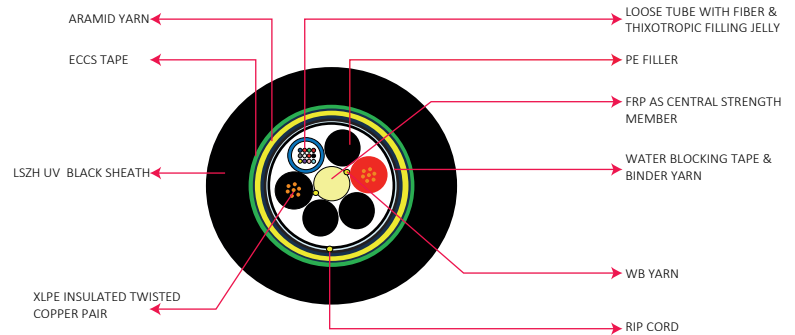
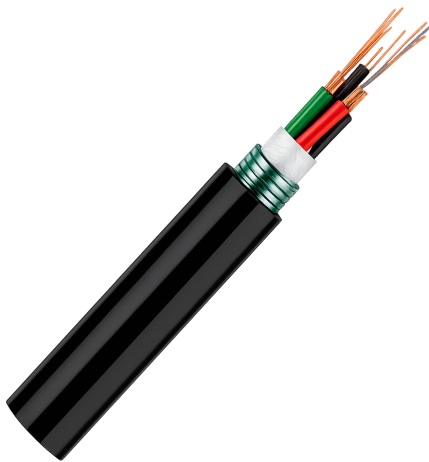


HYBRID CABLE - POWER OVER FIBER



DESCRIPTION

DATAKOM's innovative hybrid cables combine fiber optic or UTP data transmission with low-voltage power delivery in a single integrated solution. This breakthrough design eliminates the need for separate power and data cables, dramatically reducing installation complexity and cost.

Hybrid composite fiber optic cables have both fiber and copper conductors under the same cable jacket to deliver data and power to the very edge of the enterprise network.

DATAKOM Hybrid is a Class 3-rated composite cable that supports low-voltage (Class 2, 57 VDC/100 V) as well as bulk power solutions. Hybrid cable can reach distances of over 2,000 feet, this cabling option is also ideal for long-reach or remote applications, such as security cameras in parking areas or outdoor, campus-wide Wi-Fi etc.

APPLICATION

- Power cable suitable for power networks.
- Direct burial.
- Outdoors
- indoors and in cable ducts.

FEATURES

- Conductor Diameter: 14 AWG, Class 2 Copper - (As per IEC 60228 Standard).
- XLPE Insulated Core: 2.9 ± 0.1 mm overall Diameter (As per BS EN 60811-201 & BS EN 60811-501).
- Operating Voltage Under Test Condition (100/250/500/1000 Mtr) : 56/120/230 VDC.
- Operating Current Under Test Condition (100/250/500/1000 Mtr) : 1/2/5/10 Ampere
- Conductor Resistance: 8.79 Ohm/Km
- Conductors: Plain annealed stranded Copper
- Insulation: Polyethylene (XLPE) as per BS 5467 standards.
- Voltage: 600/1000V as per Standard IEC 60502
- Operating temperature: Maximum 90°C as per BS 5467 standards.
- Core identification: Two Core - Red & Black.

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CONSTRUCTION DETAILS

No of Fibre/Tube	a)04F , b)06F , c)08F - per Tube
Loose Tube	PBT Loose Tube Filled With Thixotropic Jelly (2.9 ± 0.1 mm).
No of LT For G 652 D Fibre	01 No Loose Tube (Blue)
No of Insulated Copper Pair & Wire Gauge	2 -4 , 22 - 14 AWG
Central Strength Member	Fiber Reinforced Plastic - FRP (Non Metallic) ($3.0 \text{ mm} \pm 0.1 \text{ mm}$)
Water Blocking Elements	Water Blocking Yarns & Water Swellable Tape
Core Wrapping	Water Swellable Tape With Binder Yarn
Strength Member	Aramid Yarn
Rip Cord	Rip cord provided Below Armour
Armouring	Corrugated ECCS Tape Armour
Outer Jacket / Thickness	LSZH UV Black Colour / 1.80 mm (Nominal)

FIBER CHARACTERISTICS

Attenuation (Transmission Characteristics)	
@ 1310 nm	≤ 0.36 (dB/Km)
@ 1550 nm	≤ 0.23 (dB/Km)

DISPERSION SLOPE & WAVE LENGTH

Zero Dispersion Wavelength	1300-1324 nm
Zero Dispersion Slope	≤ 0.092 ps/nm ² .km

CUT OFF WAVELENGTH

Fibre cut-off Wavelength	≤ 1320 nm
Cable Cut-off Wavelength	≤ 1260 nm

POLARIZATION MODE DISPERSION 1310 & 1550nm

At Fibre Stage	≤ 0.2 ps/sqrt.km
At Cable Stage	≤ 0.3 ps/sqrt.km

GEOMETRICAL CHARACTERISTICS

Mode Field Diameter @ 1310 nm	9.2 ± 0.4 μm
Mode Field Diameter @ 1550 nm	10.4 ± 0.5 μm
Cladding Diameter	125 ± 0.7 μm
Cladding Non Circularity	$\leq 1\%$
Core Clad Concentricity Error	≤ 0.5 μm
Coating Diameter	245 ± 10 μm
Coating/Cladding Concentricity	≤ 12 μm

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CABLE MECHANICAL CHARACTERISTICS

Max. Tensile Strength (IEC 60794-1-21-E1)	3000 Newton
Crush Resistance (IEC 60794-1-21-E3)	2000 N/10 cm
Impact Resistance (IEC 60794-1-21-E4)	25 Nm
Cable Bend Test (IEC 60794-1-2-E11)	20 X Diameter of Cable
Temperature Cycling Test (IEC 60794 1-2 F1)	-10°+85° Deg C - 12 Hrs
Flame Test Listing	NEC CMP-OF (ETL) and c(ETL)
Flame Test Method	NFPA 262
Water Penetration Test (IEC 60794 1-2 F5)	2 Meter/3 Meter/24 Hrs

CABLE PHYSICAL CHARACTERISTICS

Cable Diameter (Nominal)	11.0 mm - 15mm ± 0.5 mm
Packing Length	2.0/4.0 Km ± 5% & (SLG - 5%)
Torsion Test (IEC 60794-1-2-E7)	2 Mt/100 N/± 360 Deg/10 Cycle
Kink Test (IEC 60794-1-2-E10)	10 X Diameter of Cable

FIBER ALLOCATION SCHEME

Tube Colour	Fiber Type / Conductor	No of Fiber
Blue	SM G 652 D	XX
Red	14 AWG Ins. Copper Pair	
Black	14 AWG Ins. Copper Pair	

PART NUMBER CONFIGURATION

1	2	3	4	5	6	7	8	9	10	11
		FIBER TYPE	No.OF CORES		COPPER QTY		AWG NO:	ARMOURING	SHEATH	JACKET TYPE
DC	H	S9 =SM	04 =4CORE	C	2	W	14	EC - ECC TAPE	Z - LSZH	SJ -SINGLE JACKET
		M3 =OM3	06 =6CORE		4		16		P - PE	DJ -DOUBLE JACKET
		M4 =OM4	08 =8CORE							
			12 =12CORE							
			24 =24CORE							

** Examples

ORDERING INFORMATION

Description

**DATACOM Hybrid Cable. SM 08 Core Fiber With 2x16 AWG
Copper Wire, ECC Tape LSZH UV Protected Single Jacket -Black**

Part No.

DC-HS908C2W16ECZ-SJ

**DATACOM Hybrid Cable. SM 08 Core Fiber With 2x14 AWG
Copper Wire, ECC Tape LSZH UV Protected Single Jacket -Blackk**

DC-HS908C2W14ECZ-SJ